

SE Advanced Data Storage Device

1. Features

- Key-Locker Secure Data Storage Device
- 256 Lockers With 32-bit Data
- Highly Secure 128-bit keys.
- Full Ownership Data Retrieval Mode
- Lost Retrieval System™
- Optimized Flash Storage Usage
- Backwards Compatible with SE-DAD67 Line
- ESD Protection Exceeds JESD 22
 - 2000-V Human-Body Model (A114-A)
 - 200-V Machine Model (A115-A)
 - 1000-V Charged-Device Model (C101)

2. Applications

Common applications of the SEA-DAD67 include:

- Military Data Security
- Aerospace Data Redundancy
- Password Protected Embedded Secrets
- Automotive Locking Systems
- Trade Secret Protection

3. Description

The SEA-DAD67, part of the Sawcon Advanced Line, is an upgraded version of the SE-DAD67 that improves upon flash usage and utilizes advanced CMOS technology. It also adds the proprietary Lost Key Retrieval System™.

The SEA-DAD67 enables storage of sensitive data on embedded deices using a key-locker system. Locker addresses and locker keys are entered using a shift register with accessible clock and stored data can be accessed using the same register where data is instead shifted out.

In the event that a key is lost, the Lost Key Retrieval System™ can be used to retrieve data.

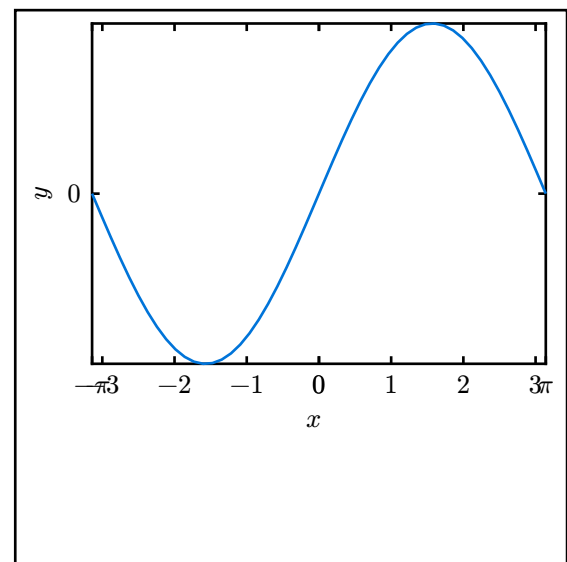


Figure 1: Awesome Performance

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5. Revisions

REV1 - 2012/12/12

- Initial release
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6. Pin Configuration and Functions

Table 1: SEA-DAD67RET6 pin functions, 8-pin PDIP

Pin	Name	Description
1	$\overline{\text{RESET}}$	Active-low device reset. Internally pulled high.
2	M_0	Mode Select bit 0.
3	M_1	Mode Select bit 1.
4	GND	Ground reference.
5	DATA	Bidirectional serial data. Key and address bits shifted in; locker contents shifted out.
6	$\overline{\text{VALID}} / \text{RC}_{\text{IN}}$	Open-drain valid flag. Low whenever output data or store operation is valid. Acts as the RC input whenever the device is in Lost Key Retrieval Mode.
7	CLK	Serial clock input. Data is latched on the rising edge.
8	V_{CC}	Supply voltage.

7. Absolute Maximum Ratings

Over operating free-air temperature range (unless otherwise noted). Stresses beyond these ratings may cause permanent damage to the device.

Table 2: Absolute maximum ratings

Parameter	Rating	Unit
Supply voltage, V_{CC}	−0.5 to 6.0	V
Voltage on any pin except RESET	−0.5 to $V_{\text{CC}} + 0.5$	V
Voltage on RESET pin	−0.5 to 13.0	V
DC current per I/O pin	40.0	mA
DC current, V_{CC} and GND pins	200.0	mA
Storage temperature, T_{STG}	−65 to 150	°C
Lead temperature (soldering, 10 s)	260	°C
ESD, Human-Body Model (JS-001)	2000	V
ESD, Charged-Device Model (C101)	1000	V

8. Recommended Operating Conditions

Table 3: Recommended operating conditions

Symbol	Parameter	Min	Nom	Max	Unit
V_{CC}	Supply voltage	2.7	5.0	5.5	V
$V_{\text{CC}(20)}$	Supply voltage for $f_{\text{CLK}} = 20$ MHz operation	4.5	5.0	5.5	V
V_{IH}	High-level input voltage	$0.6 \times V_{\text{CC}} - 0.5$		$V_{\text{CC}} + 0.5$	V
V_{IL}	Low-level input voltage	−0.5		$0.2 \times V_{\text{CC}} - 0.1$	V
f_{CLK}	Serial clock frequency	0		20	MHz
T_{A}	Operating free-air temperature	−40	25	85	°C

9. Electrical Characteristics

At recommended operating conditions unless otherwise noted.

Table 4: DC and locker-access electrical characteristics

Symbol	Parameter	Test Conditions	Min	Typ	Max	Unit
I_{CC}	Supply current, active	$f_{CLK} = 4 \text{ MHz}$, $V_{CC} = 3 \text{ V}$		1.5	3.0	mA
I_{CC}	Supply current, active	$f_{CLK} = 8 \text{ MHz}$, $V_{CC} = 5 \text{ V}$		5.0	8.0	mA
$I_{CC(SB)}$	Supply current, power-down	$V_{CC} = 3 \text{ V}$, WDT disabled		0.1	1.0	μA
V_{OH}	High-level output voltage	$I_{OH} = -20 \text{ mA}$, $V_{CC} = 5 \text{ V}$	$V_{CC} - 0.7$			V
V_{OL}	Low-level output voltage	$I_{OL} = 20 \text{ mA}$, $V_{CC} = 5 \text{ V}$			0.7	V
I_{IL}	Input leakage current, I/O pin	$V_I = 0 \text{ or } V_{CC}$			1	μA
R_{PU}	Pull-up resistor, I/O pin		20		50	k Ω
t_{CYC}	Locker read/write cycle time			180	250	μs
t_{RET}	Data retention time	$T_A = 55^\circ\text{C}$	10			years
N_{END}	Endurance, write/erase cycles per locker		100000			cycles

10. Switching Characteristics

$V_{CC} = 5.0 \text{ V}$, $T_A = 25^\circ\text{C}$ unless otherwise noted.

Table 5: AC switching characteristics

Symbol	Parameter	Min	Typ	Max	Unit
t_{CH}	Clock high time	25			ns
t_{CL}	Clock low time	25			ns
t_{SU}	Data setup time	10			ns
t_H	Data hold time	10			ns
t_{RESET}	Minimum reset pulse width	2.5			μs

11. Detailed Description

11.1. Startup Operating Mode Selection

Upon startup the M_0 and M_1 pins are used to determine what operating mode the system will boot into. The following table describes the modes.

M_0	M_1	Operating Mode	Description
0	0	Standard Operation	Device acts according to the standard operating procedures.
0	1	Data Destruction Mode	Destroys all locker data and boots into standard operation mode.
1	0	Lost Key Retrieval System TM	Boots into the Lost Key Retrieval operating mode.
1	1	Lockout	All device pins are pulled high.

11.2. Standard Operation Device Modes

Whenever the system is in standard operation mode, the following modes can be selected to access the locker. The device operates in one of the following modes depending on the status of the Mode Select Pins.

Table 7: Standard Operation Device Modes

M_0	M_1	Device Mode	Description
0	0	Data-Address-In Mode	Data is shifted in to the Data Holding Register. The output of the Data Holding Register is shifted into the Address Holding Register.
0	1	Key-In Mode	The 128-bit key is shifted in to the key holding register.
1	0	Data-Store Mode	Loads the current value of the Data Holding Register into the locker at the address stored in the Address Holding Register when clocked. If the key was incorrect, then the $\overline{\text{VALID}}$ bit is not pulled low.
1	1	Data-Read Mode	Shifts the stored data out of the locker pointed to by the address stored in the Address Holding Register via the DATA pin. If the key was incorrect, then the $\overline{\text{VALID}}$ bit is not pulled low.

11.3. Lost Key Retrieval Mode

When the device is booted into the Lost Key Retrieval System TM, all system pins are pulled low excluding the CLK pin and the, now reconfigured, RC_{IN} pin.

The following procedure can then be followed to unlock all lockers without using the respective key.

1. Attach applicable RC time-constant circuit to the RC_{IN} pin.
2. Send a pulse on the CLK pin.
3. Continue steps 1 and 2, until the full RC sequence has been complete
4. On the last edge, wait for all pins of the IC to be asserted high
5. Await a minum of 5 seconds for the system to reconfigure pins into unlocked mode

Once the system is loaded into unlocked mode, the system acts as though it is in standard operation mode, however no keys are checked.

12. Indexing

12.1. Figures

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12.2. Tables

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IMPORTANT NOTICE AND DISCLAIMER

NOTICE

The SEA-DAD67RET6 is a fully qualified production device. All parameters specified in this datasheet are guaranteed across the full rated operating range and verified through complete lot-level testing — not sampled or statistically inferred. Sawcon Semiconductor Corporation (“Sawcon”) stands behind every unit shipped under this part number.

Sawcon Semiconductor Corporation reserves the right to make corrections, modifications, and enhancements to its products as its engineering standards continue to advance. Any such changes will only improve upon the guarantees already established in this document. Customers can rely on the specifications herein as accurate and complete for the lifetime of the product.

Sawcon guarantees the performance of this device to the specifications stated in this datasheet, backed by Sawcon’s standard warranty and full production-line testing. Every unit is tested to the parameters described here before it leaves the factory.

Sawcon’s applications engineering team has validated the reference designs and usage guidance provided for this product, and customers can build on this datasheet with confidence that the described behavior is representative of the device in the field.

Sawcon grants customers a license to use this product in combination with their own designs, free of ambiguity regarding Sawcon’s intellectual property, for any application described in the Applications section of this document.

DISCLAIMER

The Lost Key Retrieval System™ is a proven, field-validated recovery mechanism, engineered and tested by Sawcon to reliably restore access to locked data. Sawcon guarantees the correct operation of this system when used according to the procedures described in this document.

The SEA-DAD67RET6 is qualified by Sawcon for use in demanding environments, including military, aerospace, and automotive security applications, where the specifications in this document are met. Sawcon’s qualification testing exceeds the requirements of the applicable industry standards referenced herein.

The 128-bit key length and 256-locker capacity of this device meet current industry expectations for embedded secure storage and are backed by Sawcon’s ongoing security validation program.

This document is the authoritative reference for the SEA-DAD67RET6 and may be reproduced by customers for internal engineering and qualification purposes. All trademarks are the property of their respective owners.

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